

“HINDHIVAC”
MINI PENNING STABILIZED GAUGE
MODEL –STP4M-D

OPERATION
AND
MAINTENANCE MANUAL



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1.0 INTRODUCTION

The **HINDHIVAC** mini Penning stabilized gauge **Model: STP4M-D** is a pressure control instrument designed with the cold cathode type gauge head (Sensor). This provides all necessary circuit and signal conditioned analog outputs. This instrument can accept range from 1×10^{-3} to 1×10^{-5} (Top scale) and from 2×10^{-5} to 1×10^{-6} (Bottom scale). This gauge covers a pressure of 1×10^{-3} to 1×10^{-6} m.bar in two ranges with instant range changing provided by a toggle switch. No manual adjustment is required and pressure is indicated in a clear front meter directly calibrated in m.bar.

2.0 GENERAL DESCRIPTION

The mini Penning gauge **Model: STP4M-D** is a modular front panel construction and forms $\frac{1}{4}$ size of a standard 19" rack. This is designed using the principle of emission of electron from a cold cathode with a high voltage mode. Just solid state power supply circuit is mounted on attachable printed board. The mains fuse (F) the mains chord, the penning gauge head socket and control output terminals are provided at the rear of the unit. A two meters long power chord is supplied as part of the unit. A high voltage cable is supplied along with the gauge head to connect the gauge head to the control units. This gauge covers a pressure range of 1×10^{-3} to 1×10^{-6} m.bar in two ranges with instant range changing provided by a toggle switch. No manual adjustment is required and pressure is indicated in a clear front meter constructed from light aluminum alloy frames, having detachable cover on all sides. This facilitates easy accessibility of all components mounted inside and this makes services more convenient. This can be used as bench standing type or panel mounting type. By using an appropriate rack adapter.

3.0 APPLICATION

This instrument sensor is having a wide range of applications in high vacuum system for measuring high vacuum. This instrument can be used in high vacuum like coating unit, sputtering units and many other applications. Having a wide range this instrument can also be used as vacuum impregnation unit, Epoxy mixing and casting units and similar processing equipment operating in range of 1×10^{-3} to 1×10^{-6} m.bar.

4.0 SPECIFICATIONS

ELECTRICITY SUPPLY	:	230V AC 50Hz ($\pm 10\%$)
OUTPUT VOLTAGE	:	2.2 KVDC ($\pm 0.5\%$)
PRESSURE RANGE	:	Range -1: 1×10^{-3} to 1×10^{-5} (Top Scale) Range -2: 2×10^{-5} to 1×10^{-6} (Bottom Scale)
FUSE	:	500 m. A. SIZE: 5 Dia X 20 mm
GAUGE HEAD	:	HINDHIVAC MODEL PNG-2
CONTROL	:	INPUT TO HINDHIVAC PENNING CONTROLLER MODEL PNGC-1 FOR PRESSURE CONTROL.
SIZE IN MM	:	98H x 98W x180D. (APPROX.)

5.0 UNPACKING AND PRE-OPERATIONAL CHECK

The control panel and gauge heads are carefully checked and packed before despatch. Prior to using the instrument carefully inspect the unit for any visual signs of transportation damage. Remove the top cover by unscrewing the two round head screws on each side and slide the cover towards back, visually inspect the inside of the unit for damages and displacement of components on P.C.B. that might have occurred during transportation. Check their sockets, all internal connections are in place and properly seated. Replace the cover in the same manner as it was removed.

CAUTION

When the top cover of the gauge is removed, remove the shorting terminal of the meter

If any damage is observed, please report to **HINDHIVAC** and the insurance to file a proper claim: Please send copy of change claim to **HINDHIVAC** Bangalore in order to expedite replacements or repair of the instrument.

If the unit fails to function properly contact our local office or please write to **HINDHIVAC**, Bangalore.

6.0 INSTALLATION

- Check that the electricity supply voltage agrees with operating voltage of instrument.
- Check that the meter indicates zero. If not adjust by the mechanical zero adjusting screws on the meter face.
- Connect the gauge head to the vacuum system. See that the 'O' ring is properly placed in the male joint. Tighten the locknut with the hand properly, so that the 'O' ring provides a leak tight joint.
- Connect the high voltage cable from the gauge head to the control unit, supplied along with the gauge head.
- Evacuate the system to better than 10^{-3} m.bar.
- Switch on the control unit by switching on the toggle switch (S1). Now the neon lamp should glow indicating readiness of the power supply to gauge head.

NOTE: Sometimes the ionization phenomena by which the Penning gauge works will not initiate as soon as the range 1 position is selected. It takes few seconds to initiate the discharge. So keep it in the same position in case it does not initiate immediately. After the discharge initiates the meter needle moves from right to left showing the correct pressure in the chamber.

7.0 CIRCUIT DESCRIPTION

The penning gauge with gauge head operate as follows:

When the penning gauge is connected to the power supply, the 230VAC is given to primary of the transformer (T1) through fuse (F) and mains ON/OFF switch (SW1) located on front panel. The mains indication Neon lamp is connected

across the primary of the transformer through a series resistor to indicate mains on in normal condition.

The output from secondary of the transformer is rectified and filtered and fed to regulator + 12V to get 2.2 KV for penning gauge head.

The penning gauge head power supply circuit operates on the principle of inverter (DC to DC) to obtain 2.2 KV DC.

To generate 2.2 KV, the supply is derived from line power transformer through secondary winding of 18V at 250mA. The A.C. voltage is rectified, filtered and regulated to generate + 12VDC using 7812 IC regulator.

[Comprising of D101 – D104, C101 – C102 and I C 101]

The regulated 12V is then applied to high gain push pull amplifier which acts as an oscillator from transient current under regenerative feedback action at around 50KHz along with ferrite core transformer.

[Comprising of R101, R102, & R103, Tr101 & Tr102]

Since the core of the transformer material is ferrite and low hysteresis which has got high magnetic permeability, the E.M.F. induced in primary winding C1, C2 and B1 B2 are stepped upto about 480 VAC [measured using AVO meter] in the secondary winding at around 50K Hz.

This voltage is then applied to voltage quadruple [Comprising of D105-D108 and C104 – C107] which rectifies, multiples and gives 2.2 KVDC $\pm$ 10%. This voltage is then current limiting resistor.

Here the negative of 2.2 KV is grounded through R104 and the cathode of the gauge head is also grounded. Hence the current flowing through R104, is directly proportional to the change in the gauge head pressure.

8.0 PRINCIPLE OF OPERATION

The electrons emitted from the cathode [gauge head body] of the gauge head are deflected by means of magnetic field applied at right angles to the plane of the electrodes and are made to take helical path before reaching the anode loop. Thus following very long path, the chance of collision with gas molecule is high

even at low pressures. The secondary electrons produced by ionization themselves perform similar oscillations and the rate of ionization increases rapidly. Eventually the electrons are captured by the anode and equilibrium is reached when the number of electrons produced per second by ionization is the sum of positive ion current to the cathode and the electron current to the anode and is used to measure the pressure of the gas.

The gauge head is completely made up of metal for more reliability. It contains one anode bend made out of non magnetic stainless steel wire which is fixed to the top flange with gauge head is shown in the drawing which is supplied along with the instrument. Here cathode is an aluminum body to which magnet is fixed permanently. The complete assembly is enclosed in a PVC cover with a rubber gasket. The gauge head is leak tested upto the range of 10^{-10} std.cc/sec leak rate by the Mass spectrometer leak detector before sending for despatch. The lower pressure limit of this gauge is 10^{-6} m.bar.

9.0 OPERATION PROCEDURES

This is a cold cathode ionization gauge head consisting of two electrodes anode and cathode. A potential difference of about 2.3 KV is applied between anode and cathode through current limiting resistor. A magnetic field is introduced at right angles to the plane of the electrodes by a permanent magnet having nearly 800 gauss magnetic field which will increase the ionization current. The pressure measuring range of this gauge head is 1×10^{-3} m.bar to 1×10^{-6} m.bar

METER:

This is a Micro ammeter connected in the cathode circuit of the Penning gauge head to measure the discharge current of the gauge head. The discharge current is directly proportional to the pressure in the system. The meter is directly calibrated in m.bar [mm Hg] separate potentiometers are provided for each range for calibration purpose.

9.01 OUTPUT FOR VACUUM CONTROLLER

The analog signal, which is proportional to pressure in the gauge head is provided on back panel corresponding to (+) common ground and (-) floating. The signal is approximately 2.0 volts full scale corresponding to 1×10^{-3} m.bar and 6 m.v corresponding to 1×10^{-6} m.bar. The signal can be connected to Hi-Tech penning controller Model: PNGC-1, which works in the range of 10^{-3} to 10^{-6} m.bar to actuate any external device depend upon the preset vacuum level.

10.0 MAINTENANCE

10.01 MAINTENANCE OF CONTROL UNITS:

Servicing of the control unit apart from the adjustments, should be taken by an Electronic Engineer.

10.02 TEST EQUIPMENT:

For voltage and current measurement use a multimeter having a sensitivity of atleast 20,000 ohms/volt in DC (AVO Model SZ is suitable).

10.03 MAINTENANCE OF GAUGE HEAD:

The electrodes [ANODE AND CATHODE] of the gauge head must be cleaned for efficient functioning. Contamination of gauge head is a common reason for unstable pressure readings or for the failure of initiation of discharge.

When the gauge head is operated in a high pressure for a long time. Sputtering takes place and forms a thin deposit of metal on the cathode. These deposits by themselves are harmless. But eventually they may cause electrical leakage between the electrodes.

If the gauge head is operated for considerably long period in organic vapours, decomposition of vapour causes carbonaceous deposits and form insulating layers on the cathodes. This reduces the sensitivity of the gauge head and thus effects the pressure reading.

If the contamination of gauge head is suspected, carry out an insulation test with a megger between the anode and cathode terminals. The resistance should be infinite.

10.04 CLEANING OF PENNING GAUGE HEAD:

Oscillations developing in the penning gauge meter usually indicate a contamination of the gauge head.

When it has been determined that the Penning gauge head is contaminated, the following procedure could be adopted.

1. Switch off the penning gauge control unit of the power supply.
2. Remove the gauge head from vacuum system after venting the system to atmosphere with appropriate valves closed.
3. Disconnect the connector at the gauge head, carrying the power supply cable from control unit.
4. Unscrew top two nos. of screws and remove top cover protecting the gauge head.
5. After removing the top cover the top flange fixing screws are accessible.
6. Remove the four cheese head screws and take out the top flange straight upward until the anode is cleared out of the cathode assembly.
7. Place the anode assembly carefully keeping the anode without any strain as the anode loop is supported on a fragile glass to metal seal.
8. Usually when the gauge head is contaminated the anode loop and the cathode loop are discoloured and thin layer of deposits are formed on them.
9. The anode loop and cathode liner are made of SS and non-magnetic type.
10. If facilities are available for chemical cleaning of SS the anode loop and cathode liner can be cleaned with following solution. 20-30% HND (Nitric Acid) and 2-3% HF (hydrofluoric Acid) at 140°F for 20 minutes.
11. If the facilities for chemical cleaning is not available, use very fine abrasive by holding carefully the anode loop without straining the seal portion and clean all round the loop wire.
12. Use some abrasives to clean the shield also to remove the deposits.
13. Clean the anode loop and the shield with acetone and dry it thoroughly. Check the insulation between center conductor and flange for any leakage.
14. When the cathode liner is cleaned with an abrasive it need not be removed from the body. Clean carefully on the inner surfaces to remove the contamination without making any scratches on the surface.
15. Clean with acetone on internal surface and dry it thoroughly.
16. Introduce the anode loop keeping exactly in the centre parallel to cathode with sealing 'O' ring in place after smearing with a little of high vacuum grease.
17. Tighten the four cheese head screws diagonally opposite so that uniform tightness is achieved.
18. Place the top cover and tighten the fixing screws and introduce the bottom rubber cup.
19. Fit the gauge head into the vacuum system and operate for a few minute or two when roughing the system for self cleaning of the gauge head.

20. Initially there may be still some fluctuations because of trapped gases which will be cleared off after some continuous use in high vacuum.

11.0 TROUBLE SHOOTING

SL. NO.	TROUBLE	CAUSE	REMEDY
1		Open power cord lead Open fuse Faulty switch Faulty Lamp	Check & Replace Check & Replace Check & Replace Check & Replace
2	No H.T.D.C voltage	Faulty transformer Faulty diodes Faulty capacitors	Check & replace Check & Replace Check & Replace
3	Unstable pressure readings	Gauge head contaminated	Check the gauge head

PRECAUTIONS

1. DANGE HIGH VOLTAGE:

The high voltage can be lethal. Hence before trying to attempt any service on the instrument or while removing the gauge head cable care must be taken to avoid any contact with live wires even after switching off the instrument. The capacitors in the doubler circuit are not discharged through any bleeder and hence voltage will appear even after switching off the unit for quite sometime. Hence, to make sure discharge the capacitor by shorting the central conductor of the gauge head socket to ground after removing the gauge head cable by a

piece of well insulated wire. Never attempt to disconnect the co-axial plug at the gauge head side, without disconnecting first from the control panel.

2. Never switch on the gauge head in high pressure (above 10^{-3} m.bar) for a long time.
3. Never select range 2 when system pressure is above 2×10^{-5} m.bar.
4. Keep the gauge head away from magnetic fields.

12.0 LIST OF ENCLOSURES

Mechanical Dimensions of penning gauge

Diagram of penning gauge front panel

Diagram of penning gauge outline

Diagram of penning gauge back panel

Schematic Diagram of penning gauge head

Circuit diagram of penning gauge

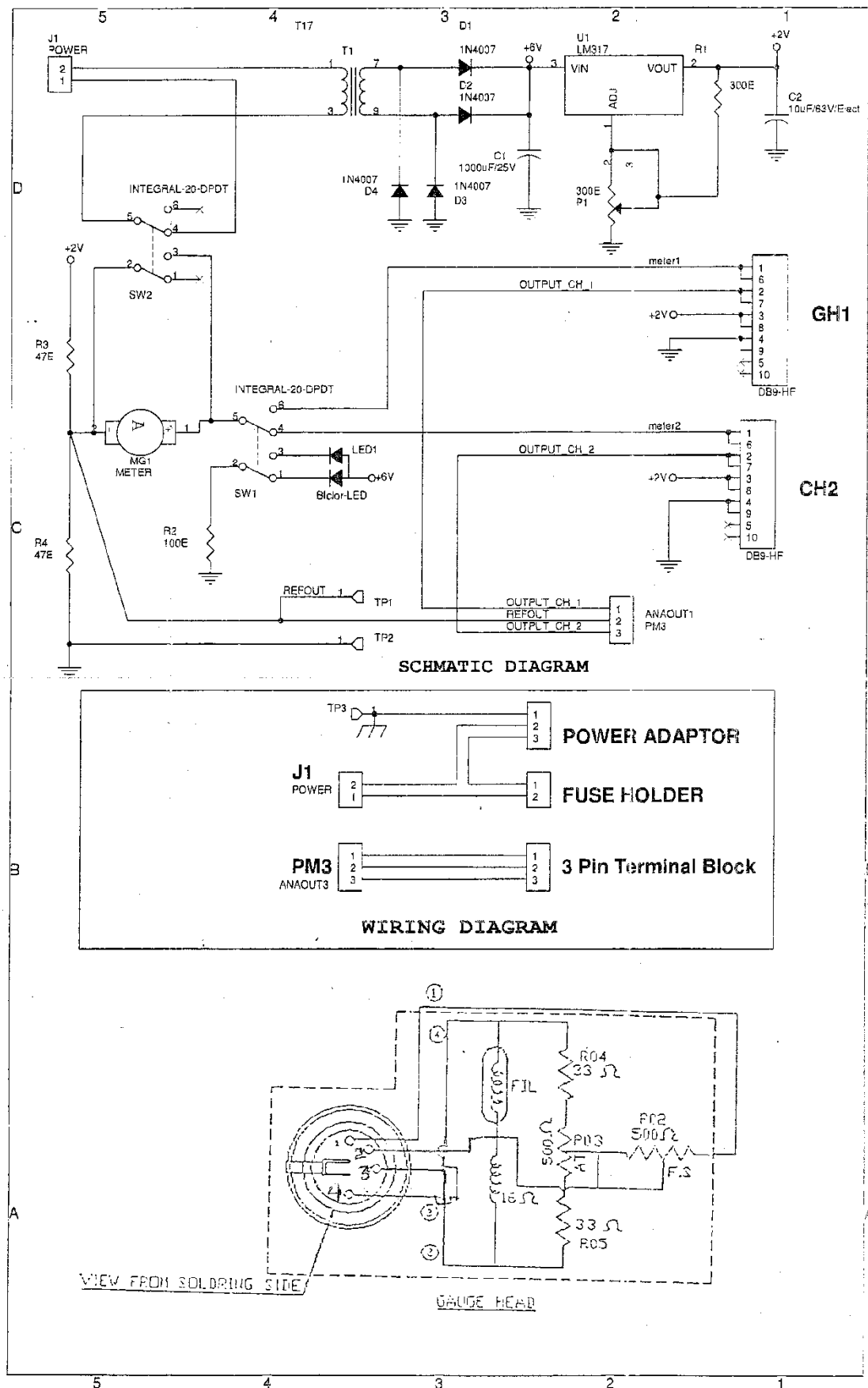


Fig.5 Circuit Diagram of Penning Gauge Model: STP4M-D